University of Chittagong
Department of Computer Science and Engineering
8th Semester B.Sc Engineering Examination 2022
Course Code: CSE-815 Course Title: Machine Learning
Full Marks: 54 Time: 4hours

[Answer any 3 questions from each section. Figures in the right hand margin indicate full marks]

Section A:

1 a) What are the differences between Artificial Intelligence (AI) vs Machine Learning (ML) vs Deep Learning (DL)? 2

b) What are the steps for solving any problem using machine learning? 3

c) Answer the provided questions using the example confusion matrix, which represents a general land cover classification. 4

		Reference			
		Developed	Low Vegetation	Forest	Water
Classification	Developed	152	35	11	13
	Low Vegetation	31	189	17	5
	Forest	2	27	231	11
	Water	7	6	8	93

i) How many total validation samples were collected?

ii) How many samples were correct?

iii) What was the overall accuracy of the classification?

iv) Measure Precision, Recall and F-1 score.

2 a) Suppose you are given the following pairs. You will simulate the k-means algorithm to identify TWO clusters in the data. 5

Data #	x	y
1	1.90	0.97
2	1.76	0.84
3	2.32	1.63
4	2.31	2.09
5	1.14	2.11
6	5.02	3.02
7	5.74	3.84
8	2.25	3.47
9	4.71	3.60
10	3.17	4.96

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Suppose you are given initial assignment cluster center as {cluster1: #1}, {cluster2: #10} – the first data point is used as the first cluster center and the 10-th as the second cluster center. Please simulate the k-means (k=2) algorithm for ONE iteration. Identify the cluster assignments after one iteration? Assume k-means uses Euclidean distance.

b) Explain the Agglomerative Hierarchical Clustering algorithm with the help of an example.

c) What are some stopping criteria for K-means clustering?

3 a) You have data from the questionnaires survey and objective testing with two attributes (acid durability and strength) to classify whether a special paper tissue is good or not. Here are four training samples.

X1= Acid Durability (seconds)	X2= Strength (kg/ square meter)	Classification
7	7	Bad
7	4	Bad
3	4	Good
1	4	Good

Now the factory produces a new paper tissue that pass laboratory test with X1= 3 and X2= 7. Apply KNN algorithm to guess what the classification of this new tissue is?

b) Compare between K-means and KNN algorithms.

c) What is AUC ROC curve? What does a ROC curve of 0.8 mean?

4 a) How does the Naive Bayes classification method perform in scenarios with correlated features? Can its assumption of feature independence compromise its accuracy in real-world datasets? What strategies can be employed to mitigate the impact of this assumption on classification outcomes?

b) In what ways does Naive Bayes classification focuses on learning the conditional probability of classes given features resemble the principles of discriminative classifiers? Can Naive Bayes be considered a hybrid approach, drawing elements from both generative and discriminative paradigms, or does it predominantly adhere to one category over the other?

c) Consider the following data set for Naive Bayesian classification problem.

Age	Income	Credit_Rating	Loan (Class)
High	Fair	Low	Risk
Low	Fair	High	No Risk
Low	Excellent	High	No Risk

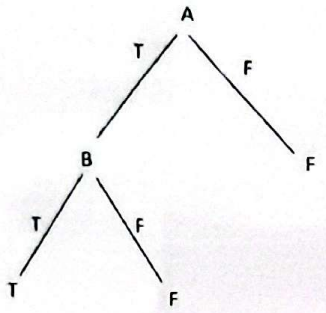
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High	Fair	High	Risk

Utilize naive Bayesian classification to forecast the class designation of a test sample (with options A=low, B=Fair, C=Low). Consider employing Laplacian correction to prevent zero probability computations if desired.

Section B:

a) Is the following figure represents the decision tree for OR function? Explain your answer. [1]



b) Consider the problem of predictiong whether the university will be closed on a particular factors which decide this are whether there is a natural disasters, whether it is a weekend or an official holiday. Suppose we have the training examples describe in following table. [4]

Disaster	Holiday	Weekend	Closed
T	T	F	F
T	T	F	T
F	T	F	F
T	T	F	F
F	F	F	F
F	F	F	T
T	F	F	T
F	F	F	T

i) What would be the effect of the Weekend attribute on the decision tree if it were made the root? Explain in terms of information gain.

ii) If you cannot make Weekend the root node, which attribute should be made the root node of the decision tree? Explain your reasoning and show your calculations. (You may use $\log_2 0.75 = -0.4$ and $\log_2 0.25 = -2$)

c) *Consider the dataset given below where T and F represent True and False respectively. What is the entropy H (Rain)?* [4]

Temperature	Cloud	Rain
Low	T	T
Low	T	T
Medium	T	F
Medium	T	T
High	T	F
High	F	F

6 (a) How does the kernel trick enhance the effectiveness of Support Vector Machines (SVMs) in handling nonlinear classification tasks? Compare and contrast the performance of linear SVMs versus kernelized SVMs on datasets with varying degrees of complexity.

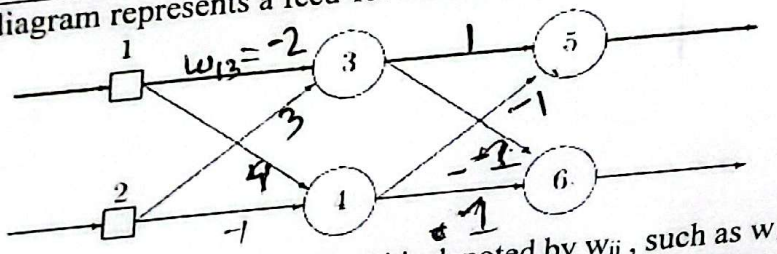
(b) Discuss the key principles behind margin maximization and the selection of support vectors in SVMs, highlighting their role in achieving robust classification boundaries. Investigate the impact of different kernel functions (e.g., polynomial, radial basis function) on the decision boundary flexibility and generalization capability of SVMs.

(c) For instance, we are given the following data points in 2D space:

x	1	4	1	2	5	5	2	6
y	1	0	-1	1	1	-1	-1	0

Classify all data points either positive or negative when $w = \{2,0\}$ and $b=-2$. What would be the distance from the separating hyperplane to any point on either positive or negative class? Finally, find the maximum margin.

7 (a) The following diagram represents a feed-forward neural network with one hidden layer:



A weight on connection between nodes i and j is denoted by w_{ij}, such as w_{13} is the weight on the connection between nodes 1 and 3. The following table lists all the weights in the network:

$w_{13} = -2$ $w_{35} = 1$ $w_{23} = 3$ $w_{45} = -1$ $w_{14} = 4$ $w_{36} = -1$ $w_{24} = -1$ $w_{46} = 1$

Each of the nodes 3, 4, 5 and 6 uses the following activation function:

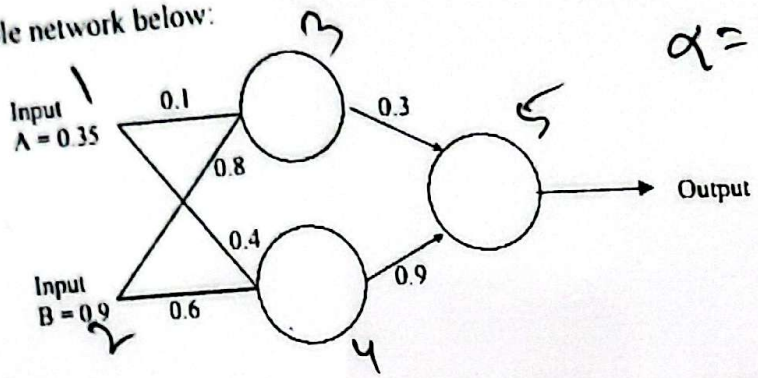
$$\varphi(v) = \begin{cases} 1 & \text{if } v \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Where v denotes the weighted sum of a node. Each of the input nodes (1 and 2) can only receive binary values (either 0 or 1). Calculate the output of the network (y_5 and y_6) for each of the input patterns:

Pattern:	P_1	P_2	P_3	P_4
Node 1:	0	1	0	1
Node 2:	0	0	1	1

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Consider the simple network below:



Assume that the neurons have a Sigmoid activation function and

(i) Perform a forward pass on the network.
(ii) Perform a reverse pass (training) once (target = 0.5).
(iii) Perform a further forward pass and comment on the result.

8 a) Explain the architecture of a convolutional neural network (CNN).

b) Consider that your input images have size 227×227×3 and filter of size 11×11×3 in the first convolution layer. In the case of AlexNet we have in the first convolution layer a total of 96 filters, a stride of 4 and a padding of 0.

i. Determine the width/height of the output of this first convolution layer.
ii. Determine the number of units within the first convolutional layer.
iii. Determine the number of trainable parameters/weights within the convolutional layer with weight sharing
iv. How many parameters would we have if we had used a FF layer with 256 units instead of a convolutional layer?

c) Differentiate between Federated learning and Incremental learning.

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